Summary

The morphological concepts and techniques introduced in this chapter constitute a powerful set of tools for extracting features of interest in an image. One of the most appealing aspects of morphological image processing is the extensive set-theoretical foundation from which morphological techniques have evolved. A significant advantage in terms of implementation is the fact that dilation and erosion are primitive operations that are the basis for a broad class of morphological algorithms. As shown in the following chapter, morphology can be used as the basis for developing image segmentation procedures with a wide range of applications. As discussed in Chapter 11, morphological techniques also play a major role in procedures for image description.

References and Further Reading

The book by Serra [1982] is a fundamental reference on morphological image processing. See also Serra [1988], Giardina and Dougherty [1988], and Haralick and Shapiro [1992]. Additional early references relevant to our discussion include Blum [1967], Lantuéjoul [1980], Maragos [1987], and Haralick et al. [1987]. For an overview of both binary and gray-scale morphology, see Basart and Gonzalez [1992] and Basart et al. [1992]. This set of references provides ample background for the material covered in Sections 9.1 through 9.4.

Important issues of implementing morphological algorithms such as the ones given in Section 9.5 and 9.6 are exemplified in the papers by Jones and Svalbe [1994], Park and Chin [1995], Sussner and Ritter [1997], Anelli et al. [1998], and Shaked and Bruckstein [1998]. Current work in the theory and applications of morphological image processing is summarized in the book by Goutsias and Bloomberg [2000], and in a special issue of *Pattern Recognition* [2000]. See also a recent compilation of references by Rosenfeld [2000]. The books by Marchand-Maillet and Sharaiha [2000] on binary image processing, and by Ritter and Wilson [2001] on image algebra, also are on interest.

Problems

See inside front cover

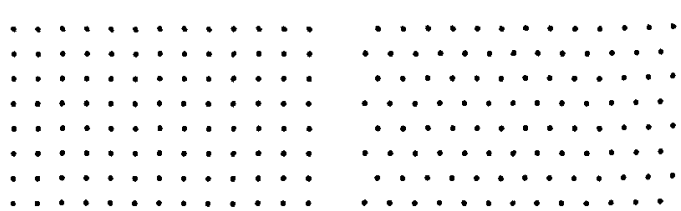
Detailed solutions to the problems marked with a star can be found in the book web site. The site also contains suggested projects based on the material in this chapter.

9.1 Digital images in this book are embedded in square grid arrangements and pixels are allowed to be 4-, 8-, or *m*-connected. However, other grid arrangements are possible. Specifically, a hexagonal grid arrangement that leads to 6-connectivity, is sometimes used (see the following figure).

(a) How would you convert an image from a square grid to a hexagonal grid?

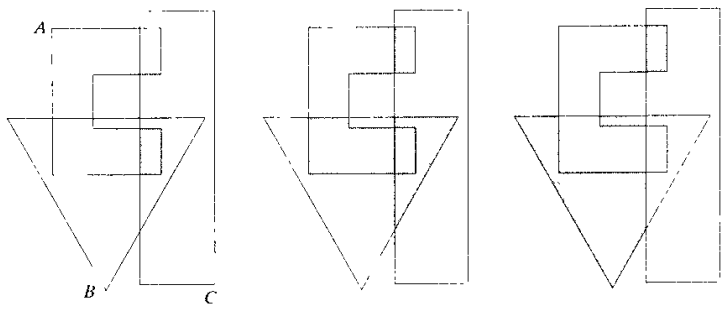
(b) Discuss the shape invariance to rotation of objects represented in a square grid as opposed to a hexagonal grid.

(c) Is it possible to have ambiguous diagonal configurations in a hexagonal grid, as is the case with 8-connectivity? (See Section 2.5.2.)



9.2 ★ **(a)** With reference to Fig. 9.1(a), sketch the set $(A \cap B) \cup (A \cup B)^c$.

(b) Give expressions for the sets shown shaded in the following figures. The shaded areas in each image constitute one set, so give an expression for each of the three images:



9.3 ★ **(a)** Give a morphological algorithm for converting an 8-connected binary boundary to an m-connected boundary (see Section 2.5.2). You may assume that the boundary is fully connected and that it is one pixel thick.

(b) Does the operation of your algorithm require more than one iteration with each structuring element? Explain your reasoning.

(c) Is the performance of your algorithm independent of the order in which the structuring elements are applied? If your answer is yes, prove it; otherwise give an example that illustrates the dependence of your procedure on the order of application of the structuring elements.

9.4 Erosion of a set A by structuring element B is a subset of A as long as the origin of B is contained by B. Give an example in which the erosion $A \ominus B$ lies outside, or partially outside, A.

9.5 The following four statements are true. Advance an argument that establishes the reason(s) for their validity. Part (a) is true in general. Parts (b) through (d) are true only for *digital* sets. To show the validity of (b) through (d), draw a discrete, square grid (as shown in Problem 9.1) and give an example for each case using sets composed of points on this grid. *Hint:* Keep the number of points in each case as small as possible while still establishing the validity of the statements.

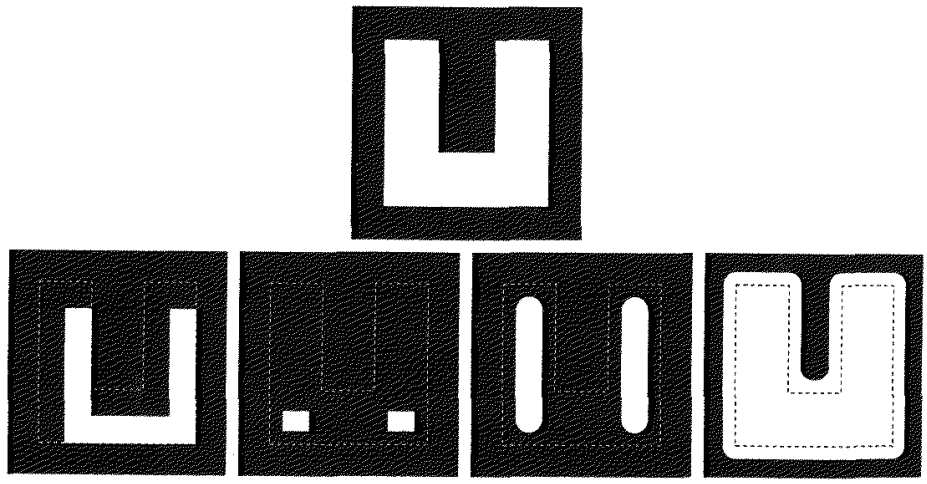
★ **(a)** The erosion of a convex set by a convex structuring element is a convex set.

★ **(b)** The dilation of a convex set by a convex structuring element is not necessarily always convex.

(c) The points in a convex digital set are not always connected.

(d) It is possible to have a set of points in which a line joining every pair of points in the set lies within the set but the set is not convex.

★**9.6** With reference to the image shown, give the structuring element and morphological operation(s) that produced each of the results shown in images (a) through (d). Show the origin of each structuring element clearly. The dashed lines show the boundary of the original set and are included only for reference. Note that in (d) all corners are rounded.



(a) (b) (c) (d)

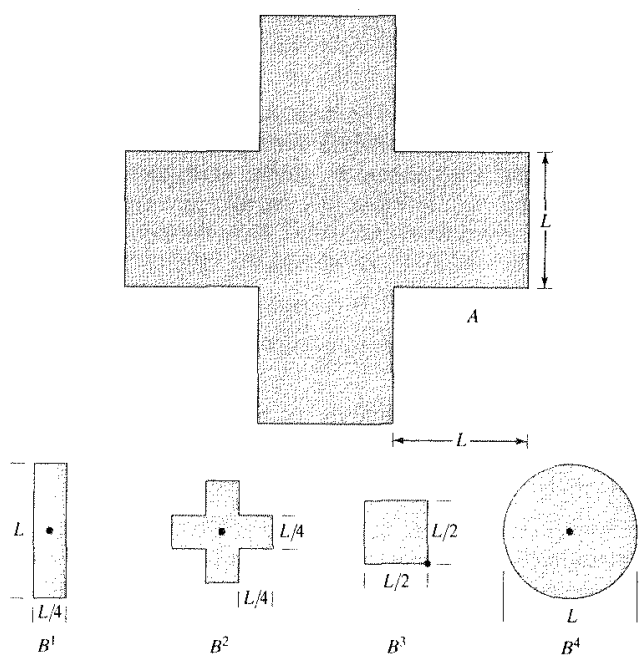
9.7 Let A denote the set shown shaded in the following figure. Refer to the structuring elements shown (the black dots denote the origin). Sketch the result of the following morphological operations:

(a) $(A \ominus B^4) \oplus B^2$

(b) $(A \ominus B^1) \oplus B^3$

(c) $(A \oplus B^1) \oplus B^3$

(d) $(A \oplus B^3) \ominus B^2$



★**9.8** **(a)** What is the limiting effect of repeatedly dilating an image? Assume that a trivial (one point) structuring element is not used.

(b) What is the smallest image from which you can start in order for your answer in part (a) to hold?

9.9 **(a)** What is the limiting effect of repeatedly eroding an image? Assume that a trivial (one point) structuring element is not used.

(b) What is the smallest image from which you can start in order for your answer in part (a) to hold?

★**9.10** An alternative definition of dilation is

$$A \oplus B = \{w \in Z^2 \mid w = a + b, \text{ for some } a \in A \text{ and } b \in B\}.$$

Show that this definition and the definition in Eq. (9.2-1) are equivalent.

9.11 **(a)** Show that the definition of dilation given in Problem 9.10 is equivalent to yet another definition of dilation:

$$A \oplus B = \bigcup_{b \in B} (A)_b$$

(This expression also is called the *Minkowsky addition* of two sets.)

(b) Show that the expression in (a) also is equivalent to the definition in Eq. (9.2-1).

★**9.12** An alternative definition of erosion is

$$A \ominus B = \{w \in Z^2 \mid w + b \in A, \text{ for every } b \in B\}.$$

Show that this definition is equivalent to the definition in Eq. (9.2-3).

9.13 **(a)** Show that the definition of erosion given in Problem 9.12 is equivalent to yet another definition of erosion:

$$A \ominus B = \bigcap_{b \in B} (A)_{-b}$$

(If $-b$ is replaced with b, this expression is called the *Minkowsky subtraction* of two sets.)

(b) Show that the expression in (a) also is equivalent to the definition in Eq. (9.2-3).

★**9.14** Prove the validity of the duality expression $(A \bullet B)^c = (A^c \circ \hat{B})$.

9.15 Prove the validity of the following expressions:

★ **(a)** $A \circ B$ is a subset (subimage) of A.

(b) If C is a subset of D, then $C \circ B$ is a subset of $D \circ B$.

(c) $(A \circ B) \circ B = A \circ B$.

9.16 Prove the validity of the following expressions (assume that the origin of B is contained in B and that Problem 9.16(a) is true):

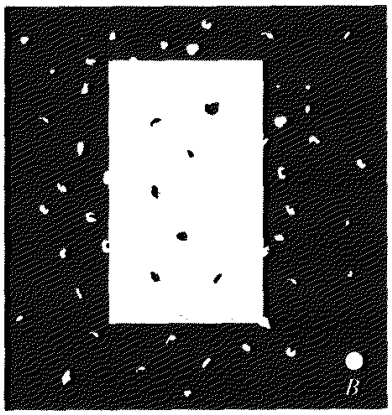
(a) A is a subset (subimage) of $A \bullet B$.

(b) If C is a subset of D, then $C \bullet B$ is a subset of $D \bullet B$.

(c) $(A \bullet B) \bullet B = A \bullet B$.

9.17 Refer to the image and structuring element shown. Sketch what the sets C, D, E, and F would look like in the following sequence of operations: $C = A \ominus B$; $D = C \oplus B$; $E = D \oplus B$; and $F = E \ominus B$. The initial set A consists of all the

image components shown in white, with the exception of the structuring element B. Note that this sequence of operations is simply the opening of A by B, followed by the closing of that opening by B. You may assume that B is just large enough to enclose each of the noise components.



★**9.18** In the example given in connection with Fig. 9.7, it is demonstrated that the squares surviving the erosion step could be fully restored by dilating them with the same structuring element used for erosion. Erosion followed by dilation is the opening of the image, and we know that, in general, opening does not result in identical reconstruction of the objects to which it is applied. Explain why full reconstruction of the remaining squares was possible in Fig. 9.7.

9.19 **(a)** Sketch the result of applying the hit-or-miss transform to the image and structuring element shown. Indicate clearly the origin and border you selected for the structuring element.

(b) Compare your result with the result of matching by correlation (Fig. 4.46), stating differences and similarities between the two.

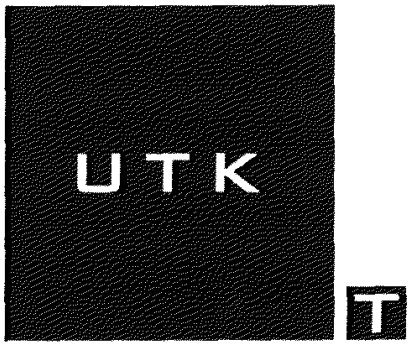
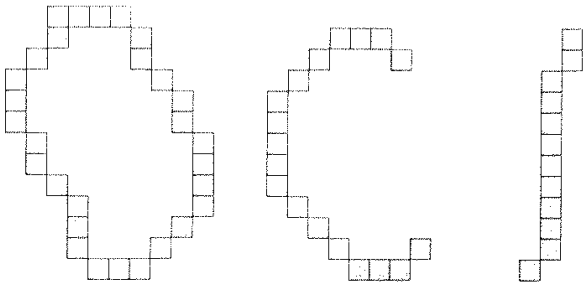


Image Structuring element

★**9.20** Three features (lake, bay, and line segment) useful for differentiating thinned objects in an image are shown in the following figures. Develop a morphological/logical algorithm for differentiating among these shapes. The input to your algorithm would be one of these three shapes. The output must be the identity of the input. You may assume that the features are 1 pixel thick and that each is fully connected. However, they can appear in any orientation.



Lake Bay Line segment

9.21 Discuss what you would expect the result to be in each of the following cases:

(a) The starting point of the region filling algorithm of section 9.5.2 is a point on the boundary of the object.

(b) The starting point in the region filling algorithm is outside of the boundary.

(c) Sketch what the convex hull of the figure in Problem 9.7 would look like as computed with the algorithm given in Section 9.5.4. Assume that $L = 3$ pixels.

★**9.22** **(a)** Discuss the effect of using the structuring element in Fig. 9.15(c) for boundary extraction, instead of the one shown in Fig. 9.13(b).

(b) What would be the effect of using a 3×3 structuring element composed of all 1's in the region filling algorithm of Eq. (9.5-2), instead of the structuring element shown in Fig. 9.15(c)?

9.23 Propose a method for fully automating the example shown in Fig. 9.16. This implies determining which black points are background points and which are points contained within the spheres (i.e., black regions completely contained within white regions). Assume that binary 1's are represented in white.

★**9.24** The algorithm given in Section 9.5.3 for extracting connected components requires that a point be known in each connected component in order to extract them all. Suppose that you are given a binary image containing an arbitrary (unknown) number of connected components. Propose a completely automated procedure for extracting all connected components. Assume that points belonging to connected components are labeled 1 and background points are labeled 0.

9.25 Suppose that the image $f(x, y)$ and structuring element $b(x, y)$ in Eq. (9.6-1) both are rectangular, with domains D_f and D_b denoted $([F_{x1}, F_{x2}], [F_{y1}, F_{y2}])$ and $([B_{x1}, B_{x2}], [B_{y1}, B_{y2}])$ respectively. For example, the closed intervals $[F_{x1}, F_{x2}]$ and $[F_{y1}, F_{y2}]$ are the ranges of x and y in the x- and y-axes of the xy-plane where the function $f(x, y)$ is defined.

★ **(a)** Assume that $(x, y) \in D_b$, and derive expressions for the intervals over which the displacement variables s and t can range to satisfy Eq. (9.6-1). These intervals in the s- and t-axes define the rectangular domain of $(f \oplus b)(s, t)$ in the st-plane.

(b) Repeat for erosion, as defined in Eq. (9.6-2).

9.26 A gray-scale image, $f(x, y)$, is corrupted by nonoverlapping noise spikes that can be modeled as small, cylindrical artifacts of radii $R_{\min} \leq r \leq R_{\max}$ and amplitude $A_{\min} \leq a \leq A_{\max}$.

★ **(a)** Develop a morphological filtering approach for cleaning up the image.

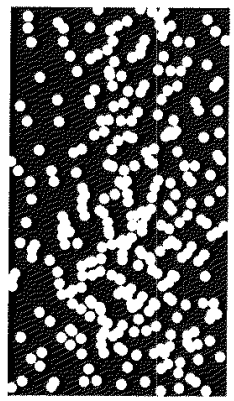
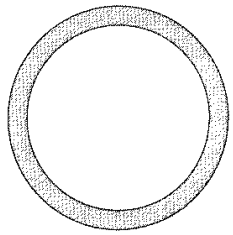
(b) Repeat (a), but now assume that there is overlapping of, at most, four noise spikes.

9.27 A preprocessing step in an application of microscopy is concerned with the issue of isolating individual round particles from similar particles that overlap in groups of two or more particles (see following image). Assuming that all particles are of the same size, propose a morphological algorithm that produces three images consisting respectively of

★ **(a)** Only of particles that have merged with the boundary of the image.

(b) Only overlapping particles.

(c) Only nonoverlapping particles.



9.28 A high-technology manufacturing plant wins a government contract to manufacture high-precision washers of the form shown in the following figure. The terms of the contract require that the shape of all washers be inspected by an imaging system. In this context, shape inspection refers to deviations from round on the inner and outer edges of the washers. You may assume the following: (1) A "golden" (perfect with respect to the problem) image of an acceptable washer is available; and (2) the imaging and positioning systems ultimately used in the system will have an accuracy high enough to allow you to ignore errors due to digitalization and positioning. You are hired as a consultant to help specify the visual inspection part of the system. Propose a solution based on morphological/logic operations. Your answer should be in the form of a block diagram.